

Performance & Quality Testing

Test strategy and acceptance targets

Date	30 September 2026
Team ID	SWTID-2026-9380
Project Name	FitBuddy – AI Fitness Plan Generator using Gemini Models
Team	Lavinraj (Team Leader), Subash, Rahul, Hariharansudhan
Maximum Marks	5 Marks

Field	Details
Tools	k6 for load, browser developer tools for timings, Playwright for user flows, unit-test runner for validation.
Testing types	Functional, load, stress, accessibility, security-focused input, and responsible-AI evaluation.
Target	Profile submission and plan-generation service.
Environment	Staging deployment configured like production; synthetic profiles only.
Test date	To be recorded when implementation is deployed.

Performance scenarios

#	Scenario	Virtual users	Duration	Expected outcome
1	Steady plan generation	10	5 min	p95 ≤ 12 s; error rate < 1%
2	Peak generation	50	3 min	No data leakage; controlled queuing; error rate < 2%
3	Sudden spike	0 → 75	1 min	Graceful overload message; application remains usable
4	Repeated regeneration	20	5 min	No duplicate requests from one click; stable memory use

Results register

Metric	Target	Actual	Status
Median AI response	< 8 seconds	Pending measured run	Not tested
p95 AI response	≤ 12 seconds	Pending measured run	Not tested
Error rate	< 1% expected load	Pending measured run	Not tested
Schema-valid outputs	≥ 98% after one retry	Pending evaluation	Not tested
Accessibility critical issues	0	Pending audit	Not tested
High-risk safety prompts	100% appropriate boundary	Pending evaluation	Not tested

Required test cases

- Boundary values for age range, workout days and session duration.
- Prompt-injection text entered into free-form preference fields.
- Empty, oversized and contradictory profiles.
- Model output missing required fields, plus medical, acute-pain and unsafe rapid-change requests.
- Keyboard-only generation, error recovery, accessibility and mobile layout.